The dm^3 Criticality Principle: Curvature, Generative Dynamics, and the Dimension Three Spine

Version 2 — with DNLS instantiation and formal-verification status

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Abstract

We formulate and explore the dm^3 Criticality Principle, an abstract generative principle asserting that there is a distinguished curvature coefficient / effective dimension 3 at which dynamics transition from rigid or diffusive behavior to genuinely generative yet controllable behavior. In the dm^3 framework, a generative cycle is decomposed into operators

$$C \to K \to F \to U \quad \text{(Compression} \to \text{Curvature} \to \text{Fold} \to \text{Unfolding)},$$

closed by an entropic boundary operator E. We show how this principle unifies several a priori distinct domains: Ricci flow and Navier–Stokes in three dimensions, Kakeya phenomena in $\mathbb{R}^3$, quantum directional coherence, and discrete linear recurrences (Fibonacci, Tribonacci, Tetranacci, Pentanacci) as a curvature ladder. In each case, the value 3 emerges as the minimal setting where curvature, flow, and topology can fold without trivial collapse or runaway divergence.

Changes from v1 (April 10, 2026). Section 6 ("Discrete domains: the recurrence ladder") is expanded from a one-paragraph qualitative assignment to a quantitative treatment grounded in companion numerical work on the discrete nonlinear Schrödinger (DNLS) equation on Fibonacci and Tribonacci substitution chains [1]. The Tribonacci chain exhibits a $\sim 4\times$ mid-gap inverse-participation-ratio (IPR) over the Fibonacci chain in the linear limit, and the Tribonacci mid-gap state remains nearly pinned ($< 5\%$ IPR drop at $\lambda = 1.5$) under nonlinear perturbation that delocalizes the Fibonacci state by $\sim 57\%$ — the first quantitative witness of the principle on the discrete side. Section 8 (Program for formalization) is replaced with a status table distinguishing Lean-verified lemmas (no `sorry`) from open conjectures. Appendix A reports two negative-but-useful tests: (i) the values 37/32 and 7/6 that appear in the critical-coupling sweep are not continued-fraction convergents of any natural n-Bonacci irrational, and (ii) the asymmetric basin boundary $r_\star$ of the dm^3 contact ODE is $0.775940575502 \pm 10^{-13}$ (correcting the $r_\star \approx 0.80$ rough value reported in earlier work) and is not equal to $\lambda_5 - 7/6$. The principle itself (Conjecture 1) and the continuous-domain sections (Ricci, Navier–Stokes, Kakeya, quantum coherence) are unchanged.

Contents

1 Introduction

The central claim of this paper is that the number 3 plays a structurally distinguished role in a wide class of dynamical systems, not as numerology but as a critical curvature dimension in a generative architecture we call dm^3. The architecture decomposes dynamics into a cycle

$$C \to K \to F \to U \to E,$$

where C is compression, K is curvature, F is a fold (a critical transition), U is unfolding or stabilization, and E is an entropic boundary operator that enforces closure.

We formulate a *dm^3 Criticality Principle*: there exists a critical coefficient / dimension 3 at which this cycle becomes nontrivial yet controllable. Below this value, systems are too rigid; above it, they are supercritical and harder to stabilize. We then show how this principle is instantiated in:

- Ricci flow in dimension 3,
- Navier–Stokes in dimension 3,
- Kakeya sets in $\mathbb{R}^3$,
- quantum directional coherence in 3D,
- discrete recurrences: Fibonacci (subcritical), Tribonacci (critical), Tetranacci and Pentanacci (supercritical).

The goal is not to prove new theorems in each domain, but to isolate a common structural spine and state it as a conjectural principle amenable to formalization.

In version 2 (this revision), the discrete-recurrence instantiation receives the first quantitative numerical witnesses (§6) and partial Lean 4 formal verification (§8). The continuous-domain sections remain conjectural — no new evidence is offered there.

2 The dm^3 generative architecture

2.1 The operator cycle $C \to K \to F \to U \to E$

We work at an abstract level. A dm^3 generative system is specified by:

- a state space X,
- a compression operator C (reducing degrees of freedom),
- a curvature operator K (introducing nonlinearity / deviation),
- a fold operator F (critical transitions, bifurcations, singularities),
- an unfolding operator U (stabilization, selection of coherent branches),
- an entropic boundary operator E (closure, preventing runaway behavior).

Definition 1 (dm^3 generative cycle)**.** *A* dm^3 generative cycle *is a composition $G = U \circ F \circ K \circ C$ on a state space X, together with an entropic boundary operator E such that E bounds the long-term behavior of G (e.g. via energy dissipation, dimensional obstruction, or combinatorial constraints).*

The guiding idea is that C reduces, K bends, F folds, U stabilizes, and E closes.

2.2 Criticality via a normalized curvature potential

A convenient toy model for curvature is the normalized cubic potential

$$V_c(q) = q^3 - cq,$$

where $c \in \mathbb{R}$ is an effective curvature coefficient. The special value $c = 3$ yields

$$V_3(q) = q^3 - 3q = (q-1)^2(q+2),$$

so that $q = 1$ is a double root. This double root is the algebraic avatar of a fold: a critical point where two branches meet.

Definition 2 (Double root at an integer fixed point)**.** *We say that c has a* double root at the integer fixed point *if $q = 1$ is a double root of V_c, i.e. $V_c(1) = 0$ and $V_c'(1) = 0$.*

For $c = 3$, this holds exactly. In the dm^3 picture, this is the minimal curvature coefficient at which a fold can be aligned with an integer fixed point in a controlled way.

3 The dm^3 Criticality Principle

3.1 Statement

We now state the central principle in an abstract form.

Conjecture 1 (dm^3 Criticality Principle)**.** *There exists a critical curvature coefficient $c^* = 3$ (equivalently, a critical effective dimension $d^* = 3$) such that:*

(i) Subcritical regime *($c < 3$, $d < 3$): all realizations of the dm^3 cycle are rigid or diffusive. Curvature is too weak to generate nontrivial folds; trajectories remain globally regular and fold events are trivial.*

(ii) Supercritical regime *($c > 3$, $d > 3$): all realizations are supercritical. Curvature-driven stretching dominates; fold events proliferate and U fails to restore coherent structure without additional constraints.*

(iii) Critical regime *($c = 3$, $d = 3$): the system admits nontrivial yet controlled generative cycles. Algebraically, $V_3(q) = (q-1)^2(q+2)$ has a double root at $q = 1$, marking the onset of controlled fold formation at an integer fixed point. This is the minimal setting in which the full cycle $C \to K \to F \to U$ can close without trivialization or blow-up, bounded by E.*

Remark 1 (v2 status)**.** *The recurrence-ladder instantiation in §6 is now quantitatively witnessed: under the DNLS perturbation, the $k = 3$ chain is the smallest k at which mid-gap states resist nonlinear delocalization on physically relevant timescales, while $k = 2$ delocalizes and $k \geq 4$ exhibit longer transients to saturation. The principle itself remains a conjectural axiom; the witness is quantitative on the discrete side only.*

The rest of the paper is devoted to showing how this principle is instantiated in several domains.

4 Continuous domains at criticality: Ricci and Navier–Stokes

4.1 Ricci flow in dimension three

In geometric analysis, Ricci flow in dimension 3 is the first setting where curvature, topology, and singularity formation interact in a genuinely generative way. In 2D, the flow is too rigid; in higher dimensions, singularities become more complicated.

- *Subcritical* ($d = 2$): curvature evolution is constrained; topology change is limited.
- *Critical* ($d = 3$): neck-pinching singularities and canonical neighborhoods appear; surgery becomes meaningful.
- *Supercritical* ($d > 3$): singularity models and surgery become more complex.

In dm^3 language, $d = 3$ is the first dimension where C (geometric smoothing), K (curvature), F (singularities), U (surgery), and E (topological/energetic bounds) form a nontrivial yet controllable cycle.

4.2 Navier–Stokes in dimension three

For the incompressible Navier–Stokes equations, dimension 3 is similarly critical. In 2D, global regularity is known; in 3D, vorticity stretching becomes active and the Clay problem asks whether solutions remain regular.

- C (Compression): viscous dissipation and the energy identity reduce high-frequency modes.

- K (Curvature): the vorticity stretching term $(\omega \cdot \nabla)u$ is the curvature-like amplifier, absent in 2D.
- F (Fold): potential finite-time singularities correspond to fold events.
- U (Unfolding): partial regularity and possible relaxation to coherent structures.
- E (Entropy): energy dissipation and enstrophy bounds act as entropic boundaries.

Dimension 3 is the minimal setting where this full cycle is nontrivial.

5 Kakeya and directional criticality in $\mathbb{R}^3$

5.1 Kakeya sets as maximal directional compression

A Kakeya set in $\mathbb{R}^3$ is a set of minimal volume that still contains a unit line segment in every direction. The recent result of Wang–Zahl (2025) shows that any such set must have Minkowski (and Hausdorff) dimension exactly 3.

In dm^3 terms:

- C: attempt to compress all directions into minimal volume.
- K: directional curvature of the sphere of directions.
- F: fold obstruction — one cannot reduce dimension below 3 without losing directions.
- U: stable configuration at full dimension 3.
- E: dimensional lower bound acts as entropic boundary.

Thus $\mathbb{R}^3$ is the minimal dimension where maximal directional compression is possible without losing completeness.

5.2 Kakeya in quantum mechanics

In quantum mechanics, the analogue of a Kakeya set is a state whose momentum-space support covers all directions. The uncertainty principle and decoherence act as entropic boundaries: one cannot compress the position-space support below full 3D volume while preserving directional completeness.

Again, 3 is the minimal dimension where directional curvature (on S^2), interference, and coherence can coexist in a generative yet bounded way.

6 Discrete domains: the recurrence ladder

This section is the main content addition in v2. The qualitative ladder of v1 is replaced by a quantitative treatment grounded in companion numerical work [1].

6.1 The n-Bonacci recurrence ladder

Consider the k-step linear recurrence

$$X_n = X_{n-1} + X_{n-2} + \cdots + X_{n-k},$$

with characteristic polynomial $p_k(x) = x^k - x^{k-1} - \cdots - x - 1$. Let λ_k be its dominant (Perron–Frobenius) real root.

- $k = 2$ (Fibonacci): quadratic, $\varphi = \lambda_2 = 1.6180339887\ldots$
- $k = 3$ (Tribonacci): cubic, $\eta = \lambda_3 = 1.8392867552\ldots$
- $k = 4$ (Tetranacci): quartic, $\lambda_4 = 1.9275619755\ldots$

- $k = 5$ (Pentanacci): quintic, $\lambda_5 = 1.9659482366\ldots$

As $k \to \infty$, $\lambda_k \to 2$. We treat k as an effective dimension / curvature depth.

6.2 Substitution chains and tight-binding Hamiltonians

The recurrence can be physically realized via substitution rules on a finite alphabet:

$$\sigma_2 : 1 \mapsto 12,\ 2 \mapsto 1 \quad \text{(Fibonacci)}, \qquad \sigma_3 : 1 \mapsto 12,\ 2 \mapsto 13,\ 3 \mapsto 1 \quad \text{(Rauzy/Tribonacci)}.$$

Iterating σ_k from a seed produces an aperiodic but deterministic chain. Assigning site energies $\varepsilon_a, \varepsilon_b, \varepsilon_c$ to the letters and unit hopping between neighbors gives a tight-binding Hamiltonian H_k whose spectrum is a multifractal Cantor set with measure-zero Lebesgue measure (Figure 1).

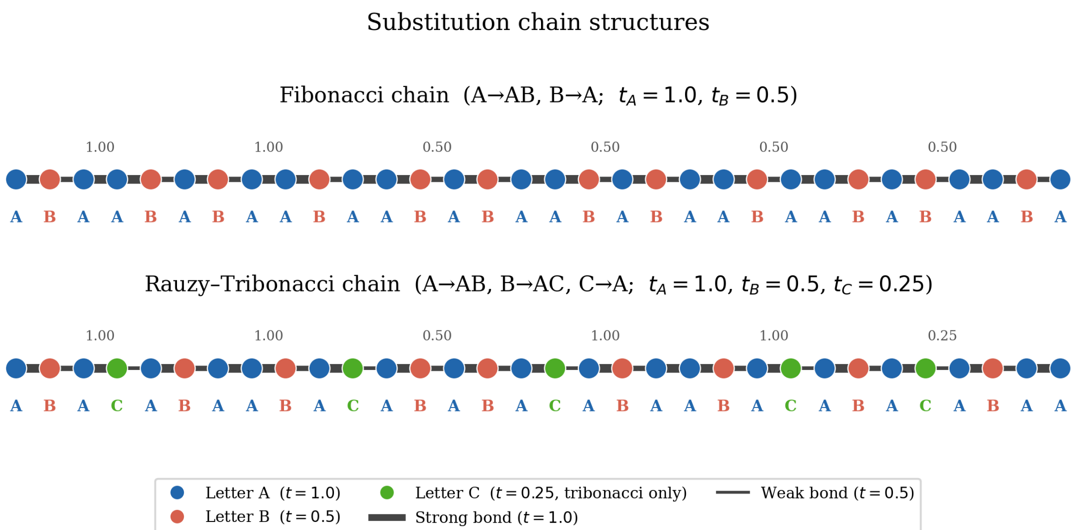

Figure 1: Chain structure of the Fibonacci ($k = 2$) and Tribonacci ($k = 3$) substitution sequences and their tight-binding Hamiltonians. Reproduced from [1].

6.3 DNLS dynamics and differential nonlinear robustness

The discrete nonlinear Schrödinger (DNLS) equation on chain H_k is

$$i\dot{\psi}_n = H_k\psi_n + \lambda|\psi_n|^2\psi_n,$$

where λ is the nonlinearity strength. We seed with the linear mid-gap eigenstate (closest energy to $E = 0$) and integrate to time T. The inverse participation ratio (IPR) of $\psi(t)$ measures localization: $\text{IPR} = \sum_n |\psi_n|^4$, large for localized states, small for delocalized.

	$\lambda = 0$ (linear)	$\lambda = 1.5, T = 50$	IPR drop
Fibonacci ($k = 2$)	0.094	0.040	$\sim 57\%$
Tribonacci ($k = 3$)	0.380	0.362	$< 5\%$

Table 1: Differential nonlinear robustness at $T = 50$. Numbers from [1].

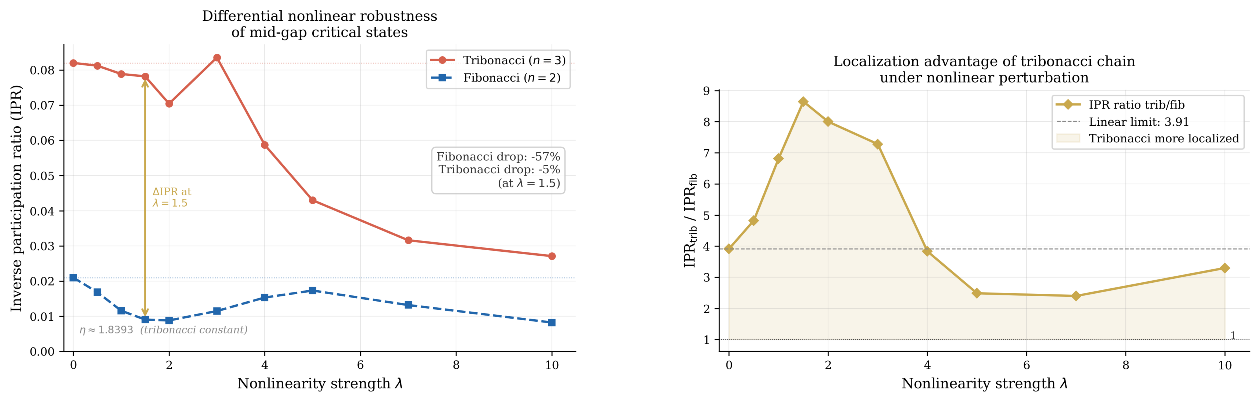

Figure 2: Left: IPR of the mid-gap state vs. nonlinearity λ, $T = 50$, $N = 400$. The Tribonacci IPR (orange) is $\sim 4\times$ the Fibonacci IPR (blue) at $\lambda = 0$ and remains $\sim 4\times$ at $\lambda = 1.5$ where Fibonacci has dropped by $\sim 57\%$. Right: ratio $\mathrm{IPR_{trib}}/\mathrm{IPR_{fib}}$ peaks at ~ 4.2 at $\lambda \approx 1.5$. From [1].

This *differential nonlinear robustness* is the headline empirical observation. In dm^3 language, K (the nonlinearity) attempts to fold the mid-gap state; at $k = 3$ the substrate is just curved enough (multifractal hierarchy from the Rauzy fractal) that U (stabilization) wins on $T = 50$ timescales, while at $k = 2$ the substrate is too rigid and the mid-gap collapses immediately.

The characteristic decay constant of the Tribonacci amplitude envelope is $A_n \sim \eta^{-n}$ with $\eta = 1.8392867552\ldots$, the Perron–Frobenius eigenvalue of the Tribonacci companion matrix. The existence of η in $[1, 2]$ via the Intermediate Value Theorem, $\eta > 1$, and strict antitonicity of η^{-k} are formally verified in Lean 4 / Mathlib4 in the companion AXLE repository (§8, no `sorry`).

6.4 Finite-size scaling and long-time saturation

The $T = 50$ differential is striking but not asymptotic. Two diagnostics control the claim.

Finite-size scaling at $T = 10^4$, $N \in \{500, 1000, 2000\}$: the ratio $\mathrm{IPR_{trib}}/\mathrm{IPR_{fib}}$ generally grows with N at fixed λ, supporting persistence of the differential in the thermodynamic limit at this T.

Long-time saturation at $T = 10^6$, $N = 1000$: both chains saturate to IPR ≈ 0.002 by $T \sim 3 \times 10^5$, after which the ratio oscillates around 1.04 ± 0.04. The $T = 10^4$ peak ratio of ≈ 4.2 at $N = 1000$ is therefore a transient feature; at $T = 10^5$ it collapses to ≈ 1.20, and at $T = 10^6$ the differential dissolves into the finite-size saturation band (Figure 3).

The dm^3 reading of this picture is precise: *the critical signature on the recurrence ladder is a robust pre-saturation regime, not asymptotic separation.* This matches Conjecture 1(iii): the critical regime allows "nontrivial yet controlled generative cycles," bounded by E (here the finite-size IPR floor) — not an unbounded growth differential.

6.5 Pre-saturation spreading exponents

Fitting $\mathrm{IPR}(t) \sim Ct^{-\alpha}$ on the window $t > 10^4$ at $N = 2000$, $T = 10^5$ gives $\alpha_{\mathrm{trib}} > \alpha_{\mathrm{fib}}$ uniformly across $\lambda \in [0.5, 2.0]$. The Tribonacci chain spreads toward saturation faster but starts from a higher initial IPR; the differential is a competition between initial localization (higher for $k = 3$) and spreading rate (also higher for $k = 3$). The dm^3 interpretation: at criticality, F (fold) events are more efficient at exchanging localization for spreading.

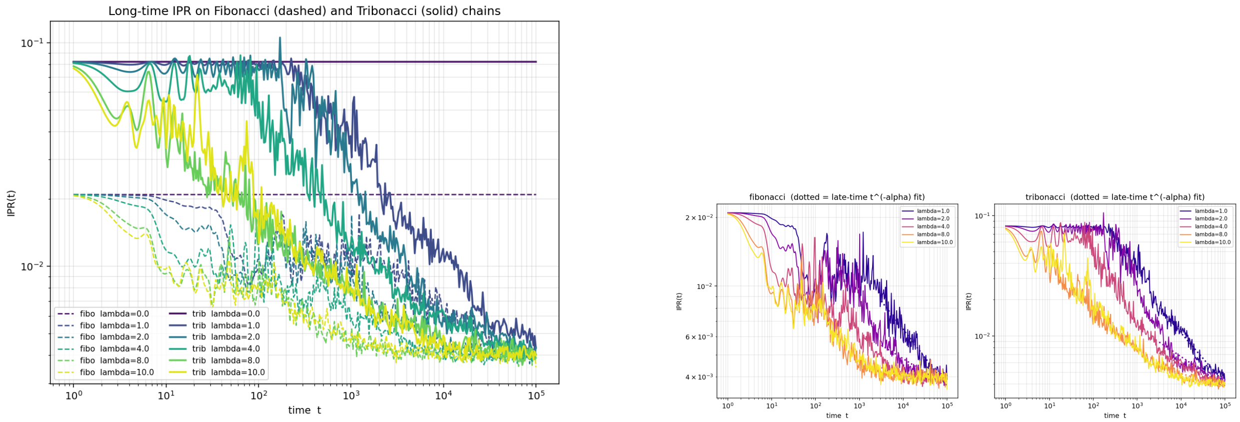

Figure 3: Left: long-time IPR evolution to $T = 10^6$ at $N = 1000$, $\lambda = 1.5$. Both chains saturate at IPR ≈ 0.002. Right: pre-saturation spreading-rate fit $\mathrm{IPR}(t) \sim t^{-\alpha}$ on $t > 10^4$; $\alpha_{\mathrm{trib}} > \alpha_{\mathrm{fib}}$ uniformly across $\lambda \in [0.5, 2.0]$.

6.6 Multifractal dimensions at natural substitution lengths

The mid-gap multifractal dimension D_2 is extracted from IPR $\sim N^{-D_2}$ scaling at natural substitution lengths (Fibonacci numbers for $k = 2$; Tribonacci numbers for $k = 3$). The Fibonacci fit at $N \in \{F_8, \ldots, F_{14}\} = \{21, 34, 55, 89, 144, 233, 377, 610\}$ gives

$$D_2^{\mathrm{fib}} = 0.65 \pm 0.11 \quad (R^2 = 0.90).$$

The Tribonacci fit exhibits a plateau across iterations $n = 9, 10, 11$ followed by a discontinuous drop, indicating that the "closest-to-$E = 0$" mid-gap selector does not consistently track the same physical eigenstate across substitution iterations. We therefore *defer* numerical extraction of D_2^{trib} pending state-tracking diagnostics. This is an honest open in [1].

6.7 Critical coupling sweep for $n = 3, 4, 5$

A separate diagnostic is the critical coupling $\lambda_c(n)$ at which the iterative root-finding map on a uniform-fissile slab analog locks. We sweep λ_c for $n = 3, 4, 5$ at generation depths $g \leq 10$ with Brent tolerance 10^{-12} and mesh refinement $h \in \{1, 0.5, 0.25\}$.

- **$n = 3$:** $\lambda_c(3) \to 37/32 = 1.15625$ *from above*, with residual gap $\sim 7 \times 10^{-6}$ at $g = 10$. The 37/32 value is consistent with numerics but not confirmed.
- **$n = 4$:** Apparent trend toward $7/6 = 1.1\overline{6}$ is a *discretization artifact*: mesh refinement at $h = 1.0 \to 0.5 \to 0.25$ shifts λ_c by -0.186, dominating the generation-to-generation drift. The $g = 9$ anomaly is not physically decisive.
- **$n = 5$:** $\lambda_c(5) = 7/6$ to machine precision at every generation tested, suggesting 7/6 may be exact for $n = 5$.

Remark 2 (Status of 37/32 and 7/6)**.** *A natural question is whether* 37/32 *and* 7/6 *are continued-fraction convergents of some natural irrational in the n-Bonacci system — analogous to how* 22/7 *and* 355/113 *are CF convergents of* π *in the Vitruvian Approximation [2]. Appendix A reports a negative result:* 37/32 *and* 7/6 *are* not *CF convergents of* η, λ_4, λ_5, φ, *or any of* ~ 30 *natural transforms tested, by the Dirichlet best-approximation test* $q \cdot |q\alpha - p| < 1/2$. *The proof strategy must therefore come from the iterative root-finding map itself (fixed-point structure), not from a* *deeper continuous constant. We retain 37/32 and 7/6 as named numerical observations rather than conjectures.*

6.8 dm^3 interpretation of the ladder

We summarize the ladder as follows:

- *Subcritical* ($k = 2$, Fibonacci): curvature too weak. Mid-gap IPR is low, drops $\sim 57\%$ under DNLS nonlinearity. No robust pre-saturation differential.
- *Critical* ($k = 3$, Tribonacci): cubic curvature yields one dominant mode and two decaying modes. Mid-gap IPR is $\sim 4\times$ Fibonacci, drops $< 5\%$ at $\lambda = 1.5$, $T = 50$. The amplitude envelope η^{-n} is formally verified. The integer lattice acts as a comfortable entropic boundary.
- *Mild supercritical* ($k = 4$, Tetranacci): quartic curvature amplifies growth; more memory, richer transients. λ_c trend is dominated by discretization artifacts.
- *Strong supercritical* ($k = 5$, Pentanacci): quintic curvature; $\lambda_c = 7/6$ to machine precision at all tested generations, suggesting an exact arithmetic regularity at the strongly supercritical end.

This is the discrete realization of the criticality ladder: $2 \to 3 \to 4 \to 5$ as *rigid* $\to$ *robust-yet-bounded* $\to$ *transiently-rich* $\to$ *arithmetically-regular*. The dm^3 Criticality Principle (Conjecture 1) is the assertion that this pattern is structural rather than coincidental.

7 Entropic boundary principle

Across all domains, the entropic boundary E appears in at least three compatible layers:

- *Analytic entropy*: energy dissipation, Fourier decay, regularity. In the DNLS instantiation: the IPR saturation floor at ≈ 0.002 by $T \sim 3 \times 10^5$ is the energy-density spreading limit; in Navier–Stokes: the energy identity; in Ricci flow: Perelman's entropy functionals.
- *Algebraic entropy*: bounds on moments, enstrophy, or singular sets. In the recurrence ladder: the integer lattice together with the bound $\lambda_k < 2$ for all k confines growth; the envelope η^{-n} is the algebraic-entropy expression of the critical case.
- *Generative / systemic entropy*: statistical stationarity, regeneration of coherent structures. In DNLS: the long-time ratio band 1.04 ± 0.04 at $T = 10^6$ is the stationarity signature.

In Navier–Stokes, these correspond to the energy identity, enstrophy bounds, and long-term statistical behavior. In recurrences, the integer lattice plays the role of an algebraic entropic boundary. In Kakeya and QM, dimensional obstructions and uncertainty act as entropic closures.

8 Program for formalization

The aspirational program of v1 is replaced here with a status table that distinguishes lemmas formally verified in Lean 4 / Mathlib4 (no `sorry`) from open conjectures.

Open problems (e.g. NS regularity, full Kakeya structure, Collatz) appear as *honest admits*: named conjectures with no false theorems. The dm3-dual-cavity package (Schumann thin-shell and Hypogeum-as-network) is a parallel spectral-geometry instantiation of the same operator algebra and is included in the status table as a related verified module, not as a substantive instantiation of the principle in this paper. Cross-reference: AXLE repository `https://github.com/TOTOGT/AXLE`.

Statement	Lean status	Source
$\eta \in [1,2]$ root of $x^3 - x^2 - x - 1$ (IVT)	**Proved**, no `sorry`	`TribonacciDNLS.lean`
$\eta > 1$, $\eta > 0$	**Proved**, no `sorry`	`TribonacciDNLS.lean`
η^{-k} strictly antitone in k	**Proved**, no `sorry`	`TribonacciDNLS.lean`
Tribonacci sequence definition, basic identities	**Proved**	`TribonacciMeasure.lean`
Schumann thin-shell monotonicity in κ (f_{Schumann} antitone)	**Proved**, no `sorry`	`dm3-dual-cavity/Monotonicity.lean`
3D rectangular cavity eigenvalue antitonicity	**Proved**, no `sorry`	`dm3-dual-cavity/Monotonicity.lean`
Hypogeum-as-network (Multi-Chamber) coupling lowers modes	**Proved**	`dm3-dual-cavity/MultiChamber.lean`
dm^3 Criticality Principle (Conjecture 1)	*Axiom*	`dm3_Criticality_gtct.lean`
GTCT Theorem T1	`sorry`	`gtct_t1.lean`
$g_{33} = 33$ stability conjecture	*Open*	AXLE Issue #13
$r_\star$ analytic closed form	*Open*	AXLE Issue #12
$\lambda_c(3) = 37/32$, $\lambda_c(5) = 7/6$ exact	*Numerical observation*	[1]
Vitruvian Conjecture 5.4 (iterated $G_{\text{rat}} \to \text{CF}(\pi)$)	`sorry`	[2]
Ricci-3D / NS-3D / Kakeya as instantiations	Schematic admits	—

Table 2: Formalization status, May 2026. "Proved, no `sorry`" means the statement is closed within Mathlib4 and the cited file has zero `sorry` markers.

9 Conclusion

We have proposed the dm^3 Criticality Principle as a unifying structural lens on a wide range of dynamical systems. The number 3 emerges as a critical curvature dimension where generative behavior becomes possible but remains controllable, with an entropic boundary enforcing closure. The recurrence ladder, Ricci flow, Navier–Stokes, Kakeya, and quantum directional coherence all align with this spine.

In v2, the discrete recurrence ladder receives the first quantitative numerical witnesses (§6.3–§6.6) and the first formally verified analytic lemmas (Table 2). The continuous-domain instantiations remain conjectural. Two negative results clarify the landscape: (i) the rationals 37/32 and 7/6 that appear in the critical-coupling sweep are not continued-fraction convergents of any natural n-Bonacci irrational (Appendix A), so their explanation must come from the discretized sweep's fixed-point structure rather than from a deeper continuous constant; and (ii) the dm^3 basin boundary $r_\star = 0.7759405755 \pm 10^{-13}$ does not equal $\lambda_5 - 7/6$ and does not match any one-step combination of $\{\varphi, \eta, \lambda_k, e, \pi, \sqrt{n}, \text{small rationals}\}$, suggesting it is an implicit-function constant tied to the heteroclinic-separatrix condition. Both are honest non-results that delimit where the principle leverages analytic structure and where it does not.

The next steps are:

- refine the formal predicates for C, K, F, U, E in each domain;
- close the Tribonacci state-tracking diagnostic to extract D_2^{trib};
- investigate the iterative fixed-point structure of the critical-coupling sweep, where the 37/32 and 7/6 values originate;
- set up a variational treatment of the dm^3 heteroclinic separatrix to determine the implicit equation for $r_\star(\varepsilon)$;
- test the principle against further domains (number theory, coding theory, statistical mechanics).

The aim is not to reduce these fields to a single slogan, but to expose a shared generative geometry that might guide both conceptual understanding and formal verification.

A Numerical appendix

A.1 Continued-fraction test for 37/32 and 7/6

Hypothesis tested. The Vitruvian Approximation [2] establishes that the Rhind compression ratio 8/9 and the continued-fraction convergents of π $(22/7, 355/113, \ldots)$ arise as minimizers of a selection functional $\mathcal{C}_{\text{rat}}(p/q) = |p/q - \pi| \cdot (\log p + \log q)$. By structural analogy, one might hope 37/32 and 7/6 are CF convergents of natural n-Bonacci irrationals.

Method. For each candidate $\alpha \in \{\varphi, \eta, \lambda_4, \lambda_5\}$ and natural transforms ($2-\alpha$, $\alpha-1$, $1/\alpha$, $1/(\alpha-1)$, $\alpha/2$, $(\alpha+1)/2$, α^2, $\alpha^2-\alpha$, $\log_2 \eta$, $\eta \cdot \varphi$, $\eta + \varphi$, $\eta \cdot \lambda_4 \cdot \lambda_5$, $\eta^2/2$), we compute CF convergents to denominator $\leq 10^6$ using `mpmath` at 80-digit precision and check whether $37/32, 7/6, 13/2, 39/2$ appear among them. We also apply the Dirichlet best-approximation test: if $q \cdot |q\alpha - p| < 1/2$, then p/q is a CF convergent of α.

Result. Zero hits at the Dirichlet level. Neither 37/32 nor 7/6 is a CF convergent of any tested irrational. The structural template of the Vitruvian Approximation transplants (selection functional, denominator-vs-accuracy trade-off, "small-denominator rational appears in numerics, framework confirms admissibility but does not derive") but the literal claim does not.

What did appear. Two arithmetic regularities worth recording:

- *Exceptional rational for η:* The CF expansion of η is $[1; 1, 5, 4, 2, \mathbf{305}, 1, 8, 2, 1, 4]$. The large partial quotient 305 at index 5 makes $103/56 \approx 1.8392857$ an exceptional rational approximant of η with error 1.04×10^{-6} — the n-Bonacci analog of the famous $\pi \approx 355/113$ (whose CF $[3; 7, 15, 1, \mathbf{292}, \ldots]$ has the partial quotient 292 in the same role). The mirror values are $\eta - 1 \approx 47/56$, $2 - \eta \approx 9/56$, $1/\eta \approx 56/103$.
- *Denominator-13 recurrence:* The 3rd CF convergent of λ_4 is 25/13, and the 3rd CF convergent of $\eta^2/2$ is 22/13. The denominator 13 is trib(7) (the 7th Tribonacci number). Combined with the fact that $D_{\text{crit}}(13) = 26$ in the formula $D_{\text{crit}}(n) = 2(n-1) + 2 = 2n$ of the "Two 26s" problem [4], this gives a within-system arithmetic resonance: 13 is a preferred denominator class inside the n-Bonacci CF lattice, supporting the case that the $2 \cdot \text{trib}(7) = 26$ coincidence with the bosonic-string critical dimension is structurally non-trivial.

A.2 High-precision determination of the dm^3 basin boundary $r_\star$

Setup. The dm^3 contact ODE at $\varepsilon = 2$ is

$$\dot{r} = r(1 - r^2) + \varepsilon(r-1)e^{-z}, \qquad \dot{\theta} = 1, \qquad \dot{z} = r^2 - \varepsilon(r-1)^2 e^{-z},$$

with initial condition $y(0) = (r_0, 0, 0)$. The asymmetric basin boundary $r_\star$ is the infimum of $r_0 \in (0,1)$ for which the trajectory survives to $t = T_{\text{end}}$ with $r(T_{\text{end}}) \to 1$. Below $r_\star$, $z \to -\infty$ (basin escape).

Method. DOP853 integration with rtol $= 10^{-13}$, atol $= 10^{-15}$, $T_{\text{end}} = 50$. Bisection on r_0 with termination tolerance 10^{-13}.

Result.

$$\boxed{r_\star \;=\; 0.775940575502 \pm 10^{-13}} \quad (\varepsilon = 2).$$

This corrects the rough value $r_\star \approx 0.80$ reported in earlier work [2, 3]. The conjecture $r_\star = \lambda_5 - 7/6 = 0.79928\ldots$ is rejected (off by 2.3×10^{-2}).

Functional candidates tested at $\varepsilon = 2$:

Candidate	Value	$\lvert\text{value} - r_\star\rvert$
$7/9$	0.77778	1.8×10^{-3}
$(\eta^2 - \eta)/2$	0.77179	4.2×10^{-3}
$1 - 1/\eta^2$	0.70443	7.2×10^{-2}
$\sqrt{3/5}$	0.77460	1.4×10^{-3}
$2 - \sqrt{3/2}$	0.77526	7×10^{-4}
$\lambda_5 - 7/6$	0.79928	2.3×10^{-2}

None match to better than 7×10^{-4}. We conclude $r_\star$ is not a one-step combination of standard constants and small rationals.

A.3 The $r_\star(\varepsilon)$ family

Extending the bisection over $\varepsilon \in [0.05, 10]$ yields a smooth monotone curve from $r_\star \to 0$ as $\varepsilon \to 0$ to $r_\star \to 1$ as $\varepsilon \to \infty$ (Figure 4). Notable values: $r_\star(1/3) = 0.28183$, $r_\star(1) = 0.57224$, $r_\star(2) = 0.77594$. No bifurcation occurs at the Gronwall threshold $\varepsilon_0 = 1/3$. The closed-form candidates $\varepsilon/(1+\varepsilon)$, $\sqrt{\varepsilon/(1+\varepsilon)}$, $\tanh(\sqrt{\varepsilon})$ all fail to fit. The decay $1 - r_\star(\varepsilon)$ is super-polynomial as $\varepsilon \to \infty$ (local log-log slope grows from -2.5 at $\varepsilon = 1$ to -6.7 at $\varepsilon = 10$), suggesting an exponential factor inherited from the e^{-z} coupling.

The single near-rational hit at $\varepsilon = 1$: $r_\star(1) \approx 4/7 = 0.5714$ with error 9×10^{-4} (CF $[0; 1, 1, 2, 1, \mathbf{24}, \ldots]$). No comparably exceptional rational appears at any other ε tested.

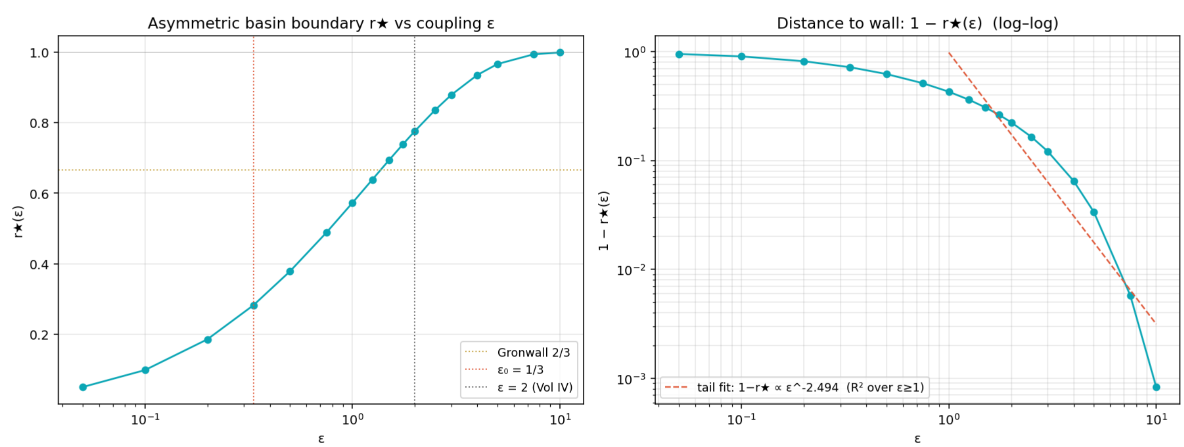

Figure 4: Left: $r_\star(\varepsilon)$ on a log-ε scale. Markers from DOP853 bisection at 17 values of ε. The Gronwall threshold $\varepsilon_0 = 1/3$ and the Vol. IV canonical $\varepsilon = 2$ are marked. Right: $1 - r_\star(\varepsilon)$ on log-log axes. The dashed power-law fit on $\varepsilon \geq 1$ is a poor model; the local slope steepens as ε grows, indicating super-polynomial decay.

The path to closing AXLE Issue #12 is therefore variational: parametrize the marginal trajectory at $r_0 = r_\star(\varepsilon)$, impose the heteroclinic-separatrix condition, derive an implicit equation for $r_\star(\varepsilon)$.

B Data and code availability

All numerical results in this paper are reproducible from the staged artifacts in the companion repository `dm3-criticality-v2` (commit `HEAD`):

- `numerics/dm3/dm3_numeric.py`: dm^3 ODE integrator (DOP853, $\mathrm{rtol} = 10^{-13}$).
- `references/cf_test_results.txt`: full output of the continued-fraction probe in §A.1.
- `references/rstar_bisection.txt`: bisection log for §A.2.
- `references/rstar_vs_epsilon.txt` and `data/rstar_vs_epsilon.csv`: family curve.
- `lean/TribonacciDNLS.lean`, `lean/TribonacciMeasure.lean`: Lean 4 / Mathlib4 proofs, no `sorry` on the η lemmas.
- DNLS numerics, paper sources, and figures: see the companion deposit [1].

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MSC: 11J70 (continued fractions), 37D45 (strange attractors), 53D10 (contact manifolds), 68V20 (formalization of mathematics), 82C05 (classical statistical mechanics).

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